Owner gauge, heat-centred full-Wick reveal, and covariance-debt frontier

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-099 reduced rational (6.5) to a lower form for the complete conditional owner

$$S_R + \frac{1}{2}\mathfrak{C}_{\text{post}} - P_R - W_0. \tag{1.1}$$

This note determines exactly which parts of (1.1) are genuine analytic content and closes one nonlinear reveal channel in the regular strict-past class.

There are four advances.

1. The complete owner is independent of its matching matrix payment. The Schur mean, posterior rebate, and paid subtraction are coordinates of one terminal Wick increment, not three positivity reserves.

2. With a matching split of both coefficient and payment, the complete owner is exactly row-additive. The R-098 cross-row posterior Schur gain is real for the bracket alone, but an equal and opposite square gap cancels it once the complete owner is restored. Finer coefficient revelation has the same exchange law: covariance mass becomes posterior-mean mass without creating net positivity.

3. The complete target has an exact posterior covariance-debt normal form. This isolates the remaining theorem as a production-weighted domination of covariance depletion by the posterior-mean current and the once-only Cameron–Martin/sextic budgets.

4. For regular mutually orthogonal strict-past no-revisit controls, a heat-centred coefficient reveal absorbs the entire future-control/Jensen residual into a full-Wick extension of R-094. This uses the signed cross sum, not an absolute frame square, and does not differentiate a selector.

The result is finite-cutoff and fixed-floor except for one explicitly labelled zero-floor reduced production-ray diagnostic. It does not prove the moving heat-baseline covariance-debt bound, rational (6.5), complete H_N, REG, arbitrary progressive/revisit H_A, $\text{OVERLAP}_{\text{src}}$, Nelson, floor removal, an interacting measure, or Sector A. Tier stays T4.

2 Complete-owner gauge collapse

Fix a conditional fibre $\mathcal{F}$. Let $B = B^T \succeq 0$ be a random Gram coefficient, let G be the derivative carrier, and let c, R, Γ be $\mathcal{F}$-measurable with $R = R^T \succ 0$. Put

$$K = \mathbb{E}_{\mathcal{F}} B, \qquad q = \mathbb{E}_{\mathcal{F}}(BG), \qquad A = K + 2R, \tag{2.1}$$

and use the R-097–R-099 coordinates

$$S_R = \frac{1}{2}\|c + A^{-1}q\|_A^2, \qquad P_R = c^T Rc, \tag{2.2}$$

$$\mathfrak{C}_{\text{post}} = \mathbb{E}_{\mathcal{F}}[B : (G^{\otimes 2} - \Gamma)] - q^T A^{-1} q. \tag{2.3}$$

Theorem 2.1 (complete-owner gauge identity)

On every such fibre,

$$\boxed{S_R + \frac{1}{2}\mathfrak{C}_{\mathrm{post}} - P_R = \frac{1}{2}\mathbb{E}_{\mathcal{F}}\left[B : ((G+c)^{\otimes 2} - \Gamma)\right].} \tag{2.4}$$

Consequently the complete owner after the once-only baseline subtraction is

$$\boxed{\mathcal{O}_{\mathcal{F}}(B,R) := S_R + \frac{1}{2}\mathfrak{C}_{\mathrm{post}} - P_R - W_0 = \frac{1}{2}\mathbb{E}_{\mathcal{F}}\left[B : ((G+c)^{\otimes 2} - \Gamma)\right] - W_0.} \tag{2.5}$$

In particular, the right side is independent of R.

Proof. Expanding the Schur square gives

$$S_R = \frac{1}{2}c^T Ac + c^T q + \frac{1}{2}q^T A^{-1}q. \tag{2.6}$$

The last term cancels the rebate in half of (2.3). Since $A = K + 2R$, subtracting P_R leaves $c^T Kc/2 + c^T q$. Adding the raw Wick term in (2.3) produces exactly the expansion of the right side of (2.4). Subtract W_0 to obtain (2.5). □

This theorem sharpens the interpretation of the R-099 payment gauge. A new choice of ridge can improve an intermediate Schur coordinate, but the matching square and actual payment change by the opposite amount. Payment tuning cannot improve the sign of the physical owner.

3 Rows and revelations exchange, but do not create mass

Let $B = \sum_a B_a$, $R = \sum_a R_a$, and $W_0 = \sum_a W_{0,a}$ be matching splits. Define

$$A_a = \mathbb{E}_{\mathcal{F}} B_a + 2R_a, \qquad q_a = \mathbb{E}_{\mathcal{F}}(B_a G), \tag{3.1}$$

$$D_{\mathrm{row}} = \sum_a q_a^T A_a^{-1} q_a - q^T A^{-1} q \geq 0. \tag{3.2}$$

The nonnegativity is the R-098 posterior superadditivity theorem.

Theorem 3.1 (exact complete-owner row additivity)

The bracket and square gaps are

$$\mathfrak{C}_{\mathrm{post}}(B;R) - \sum_a \mathfrak{C}_{\mathrm{post}}(B_a;R_a) = D_{\mathrm{row}}, \tag{3.3}$$

$$S_R(B) - \sum_a S_{R_a}(B_a) = -\frac{1}{2}D_{\mathrm{row}}. \tag{3.4}$$

Therefore

$$\boxed{\mathcal{O}_{\mathcal{F}}(B,R) = \sum_a \mathcal{O}_{\mathcal{F}}(B_a, R_a).} \tag{3.5}$$

Proof. The raw Wick part of (2.3), K, the $c^T Kc$ term, $c^T q$, the payment, and the baseline are linear under the matching split. The only nonlinear quantity is the inverse-matrix rebate. Comparing it gives (3.3), while the same comparison in (2.6) gives (3.4). Their half-weighted sum vanishes. Equivalently, apply (2.5), whose right side is visibly row-linear. □

Thus the cross-row Schur compensation is available for the stronger bracket-only target, but not as an additional reserve in the complete owner. Counting it after retaining the full square duplicates ownership. This is registered as `NG-2026-07-27-A13-COMPLETE-OWNER-CROSS-ROW-SCHUR-RESERVE`.

There is an analogous conditioning exchange. Let $\mathcal{H}_0 \subset \mathcal{H}_1$ both contain $\mathcal{F} \vee \sigma(B)$ and define

$$\mu_r = \mathbb{E}[G \mid \mathcal{H}_r], \qquad V_r = \mathrm{Cov}(G \mid \mathcal{H}_r), \qquad r = 0, 1. \tag{3.6}$$

Conditional Pythagoras gives

$$D_{\mathrm{rev}} = \mathbb{E}_{\mathcal{F}}\left[(\mu_1 - \mu_0)^T B(\mu_1 - \mu_0)\right] \geq 0, \tag{3.7}$$

$$\mathbb{E}_{\mathcal{F}}[B : V_1] - \mathbb{E}_{\mathcal{F}}[B : V_0] = -D_{\mathrm{rev}}, \tag{3.8}$$

$$\mathbb{E}_{\mathcal{F}}[(\mu_1 + c)^T B(\mu_1 + c)] - \mathbb{E}_{\mathcal{F}}[(\mu_0 + c)^T B(\mu_0 + c)] = D_{\mathrm{rev}}. \tag{3.9}$$

Finer revelation transfers exactly the same mass from covariance to mean. It creates neither a sign nor free Gaussian entropy; the information-cost boundary of R-093 remains in force.

An exact four-atom regression fixture makes both cancellations nontrivial. With equiprobable

$$G = (-2, -1, 1, 2), \quad B = (1, 4, 2, 5), \quad \Gamma = \frac{5}{2}, \quad c = \frac{2}{3}, \quad R = \frac{3}{5}, \tag{3.10}$$

and the matching split

$$B_1 = \left(\frac{1}{2}, 1, \frac{3}{2}, 2\right), \qquad R_1 = \frac{1}{5}, \qquad R_2 = \frac{2}{5}, \tag{3.11}$$

the posterior gap is 320/3927, the square gap is −160/3927, and the complete gap is zero. Refining into the first and last two atoms gives block owners −185/72 and 425/72, whose average is the coarse owner 5/3.

4 Posterior covariance-debt normal form

Let $\mathcal{H} \supset \mathcal{F} \vee \sigma(B)$, and write

$$\mu_{\mathcal{H}} = \mathbb{E}[G \mid \mathcal{H}], \qquad V_{\mathcal{H}} = \mathrm{Cov}(G \mid \mathcal{H}). \tag{4.1}$$

Theorem 4.1 (posterior covariance-debt normal form)

The complete pre-baseline owner equals

$$\boxed{S_R + \frac{1}{2}\mathfrak{C}_{\mathrm{post}} - P_R = \frac{1}{2}\mathbb{E}_{\mathcal{F}}\left[B : (V_{\mathcal{H}} - \Gamma) + (\mu_{\mathcal{H}} + c)^T B(\mu_{\mathcal{H}} + c)\right].} \tag{4.2}$$

Proof. Because B is $\mathcal{H}$-measurable,

$$\mathbb{E}[G^{\otimes 2} \mid \mathcal{H}] = V_{\mathcal{H}} + \mu_{\mathcal{H}}^{\otimes 2}. \tag{4.3}$$

Insert (4.3) into the right side of (2.4) and expand the shifted mean. □

After summing spatially and over derivative directions, define

$$\mathcal{M}_{\mathcal{H}} = \mathbb{E}\sum_i \int (\mu_{\mathcal{H},i} + c_i)^T B(\mu_{\mathcal{H},i} + c_i)\,\mathrm{d}x, \tag{4.4}$$

$$\mathcal{D}_{\mathcal{H}} = \mathbb{E}\sum_i \int B : (\Gamma_i - V_{\mathcal{H},i})\,\mathrm{d}x + 2\mathbb{E}W_0. \tag{4.5}$$

Then the target is exactly

$$\boxed{\mathcal{T} = \frac{1}{2}(\mathcal{M}_{\mathcal{H}} - \mathcal{D}_{\mathcal{H}}).} \tag{4.6}$$

Hence the missing coefficient-unconditioned theorem is equivalent to

$$\boxed{\mathcal{D}_{\mathcal{H}} \le \mathcal{M}_{\mathcal{H}} + 2\eta\mathbb{E}X + 2\zeta\mathbb{E}Y + C_{\eta,\zeta}.} \tag{4.7}$$

A stronger useful version would place $(1-\delta)\mathcal{M}_{\mathcal{H}}$ on the right, leaving a positive posterior-mean reserve. Equation (4.7), rather than positivity of $\mathfrak{C}_{\text{post}}$, is the exact frontier.

For the production frame

$$B_{\text{fr}}(Z) = \sum_\alpha M_\alpha(Z) Q_{\text{II}} M_\alpha(Z)^T, \tag{4.8}$$

the two sides have the geometric forms

$$\mathcal{M}_{\mathcal{H}} = \mathbb{E}\sum_{\alpha,i} \int \left\| Q_{\text{II}}^{1/2} M_\alpha(Z)^T (\mu_{\mathcal{H},i} + c_i) \right\|^2 \mathrm{d}x, \tag{4.9}$$

$$\mathcal{D}_{\mathcal{H}} - 2\mathbb{E}W_0 = \mathbb{E}\sum_{\alpha,i} \int \operatorname{tr}\left[Q_{\text{II}} M_\alpha(Z)^T (\Gamma_i - V_{\mathcal{H},i}) M_\alpha(Z) \right] \mathrm{d}x. \tag{4.10}$$

The R-063 forest is already the reconstruction of the raw Wick factor in this endpoint. It may not be appended a second time.

5 Heat-centred full-Wick reveal residual

Now impose the R-094 regular mutually orthogonal strict-past one-shot hypotheses. Let U_j be the Gaussian value prefix and let

$$A^{(j)} = A^0 + \sum_{k \le j} a_k, \qquad A_{>j} = A^* - A^{(j)}, \tag{5.1}$$

where $a_k \in \mathcal{F}_{k-1}$. Define

$$Z_* = U_J + A^*, \qquad Z_j = U_J + A^{(j)} = Z_* - A_{>j}. \tag{5.2}$$

For one derivative direction put

$$G_j = DU_j, \qquad Q_j = G_j^{\otimes 2} - \Gamma_j, \tag{5.3}$$

and let $d_j = Dg_j$ be the fresh derivative root. Its full Wick increment is

$$\Delta Q_j = Q_j - Q_{j-1} = 2G_{j-1} \odot d_j + (d_j^{\otimes 2} - \Delta\Gamma_j). \tag{5.4}$$

The terminal coefficient martingale and its heat-centred prefix are

$$F_j = \mathbb{E}_j B(Z_*), \tag{5.5}$$

$$L_j = \mathbb{E}_j B(Z_j) = P_{\Sigma_{j+1:J}} B(U_j + A^{(j)}), \tag{5.6}$$

and the future-control reveal residual is

$$H_j = \mathbb{E}_j[B(Z_*) - B(Z_j)]. \tag{5.7}$$

Thus $F_j = L_j + H_j$ exactly.

Theorem 5.1 (heat-centred full-Wick residual)

At finite cutoff,

$$\boxed{\mathbb{E}\langle H_j - H_{j-1}, \Delta Q_j\rangle = \mathbb{E}\langle B(Z_*) - B(Z_j), \Delta Q_j\rangle.} \tag{5.8}$$

Moreover, uniformly in the terminal cutoff,

$$\boxed{\sum_{j=j_0}^{J} \left|\mathbb{E}\int [B(Z_*) - B(Z_j)] : \Delta Q_j \,\mathrm{d}x\right| \le C_{\mathrm{full}} 2^{-j_0} X^{1/2}(1+Y)^{1/6}.} \tag{5.9}$$

Consequently, for every $\eta, \zeta \in (0,1]$,

$$\boxed{\sum_j \left|\mathbb{E}\int [B(Z_*) - B(Z_j)] : \Delta Q_j \,\mathrm{d}x\right| \le \eta X + \zeta Y + C_{\eta,\zeta},} \tag{5.10}$$

with the same Young-loss type as R-094,

$$C_{\eta,\zeta} \le C\{1 + \eta^{-1} + \eta^{-3/2}\zeta^{-1/2}\}. \tag{5.11}$$

Proof of the identity. The increment ΔQ_j is $\mathcal{F}_j$-measurable and satisfies $\mathbb{E}_{j-1}\Delta Q_j = 0$. Hence the H_{j-1} term vanishes, and the tower property gives

$$\mathbb{E}\langle H_j, \Delta Q_j\rangle = \mathbb{E}\langle B(Z_*) - B(Z_j), \Delta Q_j\rangle. \tag{5.12}$$

Proof of the estimate. R-094 assumes

$$\|d_j\|_6 \le \kappa_1 2^{j/2}, \qquad \|d_j^{\otimes 2} - \Delta\Gamma_j\|_3 \le \kappa_Q 2^j. \tag{5.13}$$

For a centred Gaussian vector of covariance C,

$$\mathbb{E}|G|^6 = (\operatorname{tr} C)^3 + 6(\operatorname{tr} C)(\operatorname{tr} C^2) + 8\operatorname{tr} C^3 \le 15(\operatorname{tr} C)^3. \tag{5.14}$$

The cumulative derivative covariance therefore gives, after absorbing the fixed low block,

$$\|G_{j-1}\|_6 \le C_G 2^{j/2}, \qquad C_G = 15^{1/6}\kappa_\Gamma^{1/2}. \tag{5.15}$$

Equations (5.4), (5.13), and (5.15) imply

$$\|\Delta Q_j\|_3 \le (2C_G\kappa_1 + \kappa_Q)2^j. \tag{5.16}$$

Apply the R-094 Gram secant

$$|B(z+a) - B(z)| \le L_B\{(1+|z|)|a| + |a|^2\} \tag{5.17}$$

with $a = A_{>j}$. The proof of R-094 (4.5)–(4.9) now applies verbatim with the constant in (5.16): the linear tail uses the strict-past L^2 control decay, and the quadratic tail uses the weighted Hardy constant $3+2\sqrt{2}$, product-space L^2–L^6 interpolation, and the accepted Littlewood–Paley sixth-moment sum. This proves (5.9). The mixed exponents 1/2 and 1/6 leave Young slack 1/3, proving (5.10)–(5.11). □

The theorem closes the nonlinear future-control reveal/Jensen residual as a signed cross. It does not control $\sum_j \|H_j\|^2$ and therefore respects the R-099 product/spike no-go. It consumes the R-094 secant allowance once. The heat baseline L_j has no standalone centering: only its complete R-077/R-097 covariance-normal row, including square, diagonal heat correction, trace, low endpoint, paid term, and R-063 forest, telescopes.

6 Adversarial boundary and reduced production diagnostic

Theorem 6.1 (abstract X/Y data do not control the debt)

For every integer $N \geq 2$, take the centred three-atom fibre

$$\Pr[(Z,G)=(N,0)] = N^{-6}, \qquad \Pr[(Z,G)=(0,\pm N^3)] = \frac{1-N^{-6}}{2}. \tag{6.1}$$

Set $B = Z^2$, $c = 0$, $W_0 = 0$, and choose any $R > 0$. Then

$$\Gamma = N^6 - 1, \quad K = N^{-4}, \quad q = 0, \tag{6.2}$$

$$\mathfrak{C}_{\text{post}} = -N^2 + N^{-4}, \qquad \mathcal{O} = -\frac{N^2}{2} + \frac{N^{-4}}{2}, \tag{6.3}$$

while the generous proxies obey

$$X = \mathbb{E}Z^2 = N^{-4}, \qquad Y = \mathbb{E}Z^6 = 1. \tag{6.4}$$

Thus no cutoff-independent bound

$$\mathcal{O} \geq -\kappa X - \zeta Y - C \tag{6.5}$$

follows from positive Gram structure, matched payment, and X/Y moments alone.

This is `NG-2026-07-27-A13-ABSTRACT-FIBRE-XY-COVARIANCE-DEBT`. It is not a production counterexample: the fibre is non-Gaussian, its Γ grows independently, and it has no spatial derivative or adapted source graph. It proves that the successor must exploit production scale-weighted covariance coupling, spatial paracomposition, or an equivalent signed Wick/forest estimate.

The production frame itself supplies a useful directional diagnostic. On the normalized one-dimensional torus, in the six-real production order, take

$$u = e_1, \qquad v = \frac{e_4 + e_6}{\sqrt{2}}, \tag{6.6}$$

$$z_0 = \xi \sin(mx + 3\pi/4)v, \qquad a = \lambda \xi^2 \cos(2mx)u, \qquad \xi \sim N(0,1). \tag{6.7}$$

The actual derivative $c = Da$ is retained. At zero floor, using

$$c_0 = \frac{3}{250P}, \qquad c_1 = \frac{243}{8000P}, \qquad \alpha = \frac{5}{9}, \tag{6.8}$$

the three linear production rows have zero quadratic coefficient, while the rational row gives

$$\mathbb{E}\Delta W = -\frac{9m^2}{400P}\lambda^2 + O(\lambda^4). \tag{6.9}$$

For $m = 2$, $P = 4$, the coefficient is $-9/400$. The Bessel H^2 proxy has quadratic coefficient

$$\frac{3}{2}(1+4m^2)^2 = \frac{867}{2}, \tag{6.10}$$

so the infinitesimal reserve ratio is only $3/57800$ and decays like m^{-2}. This ray confirms that a negative rational quadratic atom is real, but points toward spatial H^2/root-weight absorption rather than a counterexample. It omits the remaining production Gaussian roots and exact K^{-1} preimage normalization; no positive-floor or cutoff-uniform counterexample claim is made.

7 Evidence-map roadmap

The R-100 route decisions are summarized below. The complete append-only records live in `explorations/log.jsonl`; failed routes are promoted to the negative-result registry.

Route	Verdict and load-bearing reason	Successor
Tune R for positivity	Failed: (2.5) is exactly R-independent	Work on the physical owner
Count cross-row Schur gain	Failed for the complete owner: (3.3) cancels (3.4)	Split rows only by exact additivity
Refine coefficient conditioning	Boundary: covariance becomes mean with zero net change	Use revelation for analysis, not payment
Heat-centred future reveal	Advanced: (5.8)–(5.10) pay the signed full-Wick residual	Retain the complete moving heat row
PSD plus separate X/Y moments	Failed abstractly by (6.1)–(6.4)	Prove production scale-weighted coupling
Reduced production ray	Diagnostic: negative rational atom is H^2-subcritical	Seek rowwise spatial/root Wick estimate

The unique regular successor is now sharper than the R-099 target:

> Prove (4.7) for the moving heat-centred production rows, using the exact row additivity (3.5) and the R-094 full-Wick residual payment (5.10). Keep the posterior-mean current, low endpoint, diagonal heat correction, covariance trace, q/r means, actual Cameron–Martin payment, and R-063 forest once.

Only after this closes the regular complete H_N packet may REG be assembled. Arbitrary progressive/revisit H_A and $\mathrm{OVERLAP}_{\mathrm{src}}$ require the separate R-080/R-081 temporal assembly and cannot be inferred from the regular theorem.

8 Executable verification

The primary executable uses random noncommuting positive matrices, exact fractions, a 216-atom strict-past product filtration, analytic Gaussian moments, dyadic kernels, the abstract divergent fibre, and the reduced production ray. The independent executable imports no numerical or symbolic package and no primary module; it rederives the identities with hand-written two-by-two rational linear algebra and an independent exact finite filtration. The integrated verifier pins both routes, all authorities, the proof note and PDF, both negative results, exploration decisions, generated ledgers, and the honest open frontier.

Run from the repository root:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
  a13_classii_owner_gauge_heat_centered_covariance_debt_reduction_verify.py
```

9 Devil's-advocate

Objection 1: the payment gauge has proved positivity. DISMISSED. Equation (2.5) proves the opposite: matching payment is a coordinate choice, and the physical owner can have either sign.

Objection 2: R-098 posterior superadditivity supplies an extra cross-row reserve. VALID-with-mitigation. It supplies a reserve for the bracket-only target. Equation (3.4) cancels it exactly in the complete owner, and the new negative record forbids double counting.

Objection 3: finer revelation makes posterior covariance smaller and therefore improves the estimate. DISMISSED. Equations (3.8)–(3.9) show an equal mean-current increase. Net ownership is invariant, and R-093 still charges information.

Objection 4: R-094 only treated the fresh quadratic Wick tensor, so (5.9) silently changes the theorem. DISMISSED. The new proof explicitly expands the full increment in (5.4), derives its L^3 scale in (5.14)–(5.16), and only then reuses the R-094 secant/Hardy argument. This extension is independently executed.

Objection 5: (5.10) squares the nonlinear frame and contradicts R-099. DISMISSED. The estimate is for the absolute value of each expected signed cross $[B(Z_*) - B(Z_j)] : \Delta Q_j$, not for $\sum_j \|H_j\|^2$ or a pathwise frame square.

Objection 6: the abstract fibre disproves the production theorem. UPHELD against that reading. It has no production Gaussian/spatial graph and is only a method no-go. Its purpose is to prove that PSD and separate moments are insufficient inputs.

Objection 7: the reduced production ray closes the missing bound because its ratio is small. UPHELD. It omits production roots and the exact source preimage, and is at zero floor. It is a direction finder and falsifier, not a theorem for all controls.

Objection 8: the heat baseline L_j is itself a centred martingale and can be dropped. DISMISSED. Only the complete R-077/R-097 covariance-normal row centres and telescopes. Separating its coefficient fragment would again duplicate or delete heat, trace, square, low, payment, and forest owners.

Result footer

```
Result ID: A13-CLASSII-OWNER-GAUGE-HEAT-CENTERED-
           COVARIANCE-DEBT-REDUCTION
Ledger:    R-100.
Title:     Owner gauge, heat-centred full-Wick reveal, and
           covariance-debt frontier.
Precise statement:
           The complete R-099 owner is payment-gauge invariant and exactly
           row-additive; Schur row gains and revelation gains exchange with
           equal opposite square/covariance terms. The target is exactly
           posterior mean minus covariance debt. In the regular strict-past
           no-revisit class, the future-control reveal residual is bounded
           by a full-Wick R-094 secant with one X/Y allocation.
Scope:     Finite cutoff and fixed positive floor for the theorem. Regular
           mutually orthogonal strict-past one-shot controls for the
           full-Wick residual. The reduced production ray is a separately
           labelled zero-floor diagnostic only.
Dependencies:
           R-063; R-077; R-079; R-092; R-094; R-097; R-098; R-099.
Evidence grade:
           ANALYTIC, EXACT, EXECUTED, BOUNDARY.
Reproduction command:
           E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
```

a13_classii_owner_gauge_heat_centered_
covariance_debt_reduction_verify.py

Expected output:
Manifest-pinned primary, standard-library independent, integrated, and aggregate PASS; exit zero.

Negative results:
NG-2026-07-27-A13-COMPLETE-OWNER-CROSS-ROW-SCHUR-RESERVE;
NG-2026-07-27-A13-ABSTRACT-FIBRE-XY-COVARIANCE-DEBT.

Falsification gate:
Failure of owner collapse, ridge invariance, exact row or revelation exchange, covariance-debt normal form, full-Wick reveal identity/moment scale/secant estimate, exact fixtures, authority/hash/PDF/ledger/count, or T4 scope check.

Tier before / after:
T4 / T4; no promotion.

No-overclaim statement:
No moving heat-baseline production debt bound, rational (6.5), complete H_N, REG, progressive H_A, OVERLAP_src, Nelson, floor removal, measure, T5--T7, or Sector-A closure is proved.

Next required action:
Prove the rowwise scale-weighted moving heat-baseline covariance-debt bound (4.7), then close regular H_N and REG before the separate progressive/revisit assembly.